

API PIPES ERW & SEAMLESS PIPES

CHEMICAL REQUIREMENTS FOR LADLE ANALYSES, %

C ^{*5} max	Si max	Mn ^{*5} max	P max	S max	Cb min	V min	Ti min
0.21 (0.22)		0.90	0.04	0.05			
0.26 (0.27)		1.15	0.04	0.05			
0.28 (0.29) ^{*13}		1.25	0.04	0.05			
0.30 (0.31) ^{*13}		1.35	0.04	0.05			
0.28 (0.29) ^{*13}		1.25					
0.30 (0.31) ^{*13}		1.35	0.04	0.05			
0.28 (0.29) ^{*13}		1.25					
0.26		1.35	0.04	0.05	0.005 ^{*7}	0.02 ^{*7}	0.03 ^{*7}
0.26		1.35	0.04	0.05	0.005 ^{*7}	0.02 ^{*7}	0.03 ^{*7}
0.26		1.40	0.04	0.05	0.005 ^{*9}	0.02 ^{*9}	
0.23 ^{*10}		1.60 ^{*10}	0.04	0.05			
0.26 ^{*12}	0.35	1.40 ^{*12}	0.04	0.05			
0.26 ^{*12}	0.35	1.40 ^{*12}	0.04	0.05			

*8 For grades X65 in sizes 16 in. and larger with wall thickness 0.50 in. and less, the chemical composition shall be as shown or as agreed upon between the purchaser and manufacturer. For other sizes and wall thicknesses the chemical composition shall be as agreed upon between the purchaser and manufacturer. (Applicable to welded pipe only).

*9 Either columbium or vanadium or a combination of both shall be used at the discretion of the manufacturer.

*10 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible.

*11 For seamless pipe of grades X65 and X70, the chemical composition shall be agreed upon between the purchaser and manufacturer.

*12 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.50 per cent.

*13 For cold expanded seamless pipe in size 20 in. and larger, the maximum carbon content shall be 0.28 per cent.